

Short communication

Treatment of medium-sized juxtapapillary melanoma with external Co-60 photon therapy[☆]

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Abstract

Twenty-six patients with medium-sized juxtapapillary choroidal melanomas were irradiated using D-shaped Co-60 beams. The overall 5-year actuarial survival was 86% with a trend towards better 5-year local control with 70 vs. 60 Gy (100 vs. 75%). Of patients with pre-treatment $\geq 20/200$ visual acuity, 37% had functional vision at 5 years.

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1. Introduction

Choroidal melanoma is the most common primary intraocular malignancy with an annual incidence of 6–7 per million. Plaque brachytherapy [4,3] and charged-particle irradiation [8,7] provide a certain amount of visual preservation with the same disease-specific survival as enucleation [1,10]. Reports from the Collaborative Ocular Melanoma Study (COMS) have confirmed that, for medium-sized lesions, plaque brachytherapy can preserve 20/200 vision in more than half of patients at 3 years with the same 81–82% 5-year overall survival as enucleation [3,6].

Juxtapapillary lesions were excluded from the COMS trial. Because of the proximity to the optic nerve there is an increased risk of optic neuritis with plaque brachytherapy. As well, adequate tumor coverage is difficult to achieve, even with modified plaque contours. Charged-particle irradiation can provide adequate tumor dosing but significant sight-limiting treatment complications have been

documented [7]. An alternative approach using a pair of small, obliquely oriented, D-shaped Co-60 beams was proposed by Chenery et al. in 1977 [2]. The technique is simple and reproducible with good intra-tumoral and intra-ocular dose distribution. Significantly, the dose to the lens is limited to less than 10% of the tumor dose (Fig. 1).

2. Methods and materials

Between 1986 and 2000, 26 patients were treated for juxtapapillary lesions classified as medium-sized according to the revised COMS criteria (apical height 2.5–10 mm, basal diameter ≤ 16 mm). In this report, the term ‘juxtapapillary’ is used to signify a lesion ≤ 2.5 mm from the optic disc judged by the treating team to be unsuitable for plaque brachytherapy.

Initial patient evaluations included a detailed ophthalmologic exam with fundus drawings, fluorescein angiography, photography, A and B-scan ultrasonography as well as a metastatic work-up including liver enzyme profile, chest X-ray, and abdominal ultrasound.

Our standard follow-up schedule included a visit 1 month after treatment followed by a visit every 3–4 months for

[☆] Toxicity data presented at the 2001 annual meeting of the Canadian Association of Radiation Oncologists in Quebec.

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